

# Omar Abdelmotaleb

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## EDUCATION

**Stevens Institute of Technology** | Hoboken, NJ

Master of Science in Software Engineering | GPA: 4.0/4.0

May 2024

Bachelor of Science in Computer Science

May 2023

**Coursework:** Artificial Intelligence, Machine Learning, DevOps, Concurrent Programming, Algorithms, Data Structures, Database Management, IT Security, Agile Methods, Project Management, Computer Architecture, Linear Algebra, Statistics, Probability

## SKILLS

**Programming:** Python, Java, C, C++, Groovy, R, SQL, Android, Erlang, OCaml, HTML, CSS, JavaScript, XML

**Software:** Visual Studio Code, Visual Studio, IntelliJ, RStudio, Android Studio, PyCharm, Eclipse, Sublime, MS Office Suite

**Databases:** Amazon RDS, PostgreSQL, MongoDB, Firebase

**Frameworks:** Django, Spring Boot, REST

**Testing:** Mockito, JUnit, Postman, Insomnia, GoogleTest

**OS:** Windows, Ubuntu, UNIX

**DevOps:** Amazon EC2, AWS, DigitalOcean, GitHub, Git, Apache Subversion, Jenkins, Bash, Docker, Jira, Confluence

## WORK EXPERIENCE

**Software Developer Intern** | Hampton, VA

June 2022–August 2023

NASA Langley Research Center – POST2 & GRAMs

- Developed the Python and Groovy test script on Windows used for the Program to Optimize Simulated Trajectories II (POST2) debugging and refactoring all of the source code over a CI/CD pipeline with Jenkins and Apache Subversion
- Utilized a Linux server to run multiple builds on the POST2 project, completing hundreds of unit tests in C/C++ per run
- Refactored and redesigned unit tests in Visual Studio C++ for the Global Reference Atmospheric Model suite (GRAMs)
- Integrated the GoogleTest framework into the GRAMs back-end for the development team to upscale test capabilities

**Lead Software Engineer** | Hoboken, NJ

September 2022–May 2023

GridDiscovery.com

- Developed a web-based application for a startup, leading a team of 5 developers to deliver a cohesive tech solution
- Led weekly Scrum meetings with project clients for feedback and organized milestones for developers to complete in Agile
- Configured an Amazon EC2 backend server, SSL certificate, and PostgreSQL database over a private subnet in Amazon RDS to securely host data for real-time dynamic API querying with Django
- Analyzed building data through pandas library and generated API documentation for the frontend team to use with ReactJS

**Back End Developer** | Hoboken, NJ

July 2020–August 2022

NYC146

- Completed back-end Java code using Spring Boot and MongoDB on a team of 12 developers for a web application startup that allows students visiting New York City to find events around them to attend based on season and price
- Optimized 25% of the back-end services using Mockito and JUnit to debug existing code and search uncovered blocks
- Tested back-end services using Postman to send HTTP requests to the REST controller
- Created a recommendation algorithm to suggest users who to follow based on existing followers
- Designed 80% of the reservation service allowing users to reserve their slot in an event and notify when it occurs

## PROJECT EXPERIENCE

**ASL AI Image Transcriber** | Hoboken, NJ

May 2023

- Applied linear algebra with artificial intelligence and machine learning to interpret images of hand signs and translate them
- Experimented on images with various methods including eigen-decomposition, distance-based clustering algorithms, Singular Value Decomposition, neural network classifiers, and Gaussian functions for smoothing to optimize performance
- Performed in Python from scratch and using data science packages including numpy, pandas, scipy, and sklearn

**DevOps590** | Hoboken, NJ

April 2022

- Collaborated with a partner for the Stevens DevOps graduate course, developing a website on a CI/CD pipeline
- Code hosted on a GitHub repo for the HTML, CSS, and JavaScript with DigitalOcean configured for server hosting
- Jenkins pulls code on a push webhook, triggering a Bash script to deploy the website using a NGINX container on Docker

**VisitID** | Philadelphia, PA

February 2020

DragonHacks 2020

- Won **best UI/UX design** in a team of 4 during a 24-hour hackathon at Drexel University developing an Android app
- Utilized GPS for events around the user, prompting upon entering the area, speeding up attendance for the coordinator
- Developed the website using HTML, CSS, and JavaScript to allow users to create events for the application
- Implemented Firebase API to communicate event signups and attendance between the website and the application